
The EMIC Review

Student Research on Emerging Markets

Volume 1, Number 1 • Spring 2026

Emerging Markets Institute Club

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Beyond the Final Whistle: The Post-Event Economics of World Cup Hosting in Emerging Markets

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Abstract. This paper asks whether hosting the FIFA World Cup functions as a credible investment signal for emerging markets or as a case of capital misallocation. It compares three recent emerging-market hosts—South Africa (2010), Brazil (2014), and Qatar (2022)—using a longitudinal pre-post design built on GDP growth, foreign direct investment, and international tourism, read alongside case-specific qualitative evidence. All three hosts show a short-run tourism and visibility gain, but none shows a durable break in its investment trajectory, while all three carry lasting fiscal and maintenance burdens from specialised venues. The evidence sits closer to the capital-misallocation account than to the signaling one.

“Among all unimportant subjects,” said Pope John Paul II, “football is by far the most important.”

1. Introduction

In the years leading up to the 2022 World Cup, Qatar spent an estimated total of US \$220 billion to develop and build stadiums, transportation routes, tourist accommodations, and other infrastructure necessary to host the quadrennial mega-event effectively: a figure that equals the entirety of the Gulf state’s GDP in 2025.

Qatar, and other emerging economies that have previously hosted the FIFA World Cup, is willing to commit extraordinary levels of capital to the event and justify their spending by the perceived net economic benefits that are associated with hosting. These benefits stem from increased economic activity generated through broadcasting rights, international tourism, and related commercial ventures. Gains also extend beyond direct financial returns; hosting the FIFA World Cup allows nations with a rare global platform to project their culture, history, and national identity to a world-wide audience, enhancing their soft power and international visibility.

Recent emerging economy host nations of the FIFA World Cup include South Africa in 2010, Brazil in 2014, and Qatar in 2022. Each nation has invested unprecedented sums into developing infrastructure, facilities, and event operations to stage the tournament. However, the extent to which such investments yield tangible long-term economic benefits remains contested. One school of thought, investment signaling, argues that such emerging economies take advantage of the global visibility of the World Cup to project stability and modernity, attracting foreign investment following the commencement of the tournament.

Conversely, critics contend that mega-events like the FIFA World Cup result in capital misallocation and are driven by political incentives and desire for status instead of economic rationale; outcomes for emerging markets are marked by unsustainable public spending, corruption, mounting debt, and underutilized “white elephant”¹ infrastructure once the tournament concludes. This paper seeks to address whether hosting mega-sporting events like the FIFA World Cup yields long-term economic benefits or represent inefficient capital allocation for emerging economies, and to investigate the hypothesis that hosting the FIFA World Cup generates measurable net-positive financial gains to justify the scale of initial expenditure.

2. Literature Review and Theoretical Framework

2.1 Investment Signaling Theory

Investment signaling theory contests that costly and highly visible actions can create legitimacy and credibly reveal information about a government’s capacity, stability, and long-term economic prospects. When considering a state as the potential location for future foreign direct investment (FDI) projects, high-quality states can differentiate themselves by undertaking investments that would be prohibitively costly for lower-quality ones. Mega-events like the FIFA World Cup or the Olympics fit this criterion: by committing to large infrastructure upgrades, logistical coordination, and event planning on a massive scale, the host government sends a credible signal about their state capabilities and institutional quality. As these investments are sunk, publicly available, and difficult to falsify, they function as a separating signal designed to shift foreign perceptions and reduce information frictions for investors and tourists.

Prior research shows that mega-events increase international visibility and brand awareness for the host nation, altering how foreign consumers and investors perceive the host. Fourie and Santana-Gallego (2011) find that the boost in tourism in and around the event can be attributed as partly reputational; that FIFA or the International Olympic Committee (IOC) has vetted and approved a nation as qualified to host the event functions somewhat like a seal of approval. Similarly, Baade and Matheson (2004) found that the media’s role emphasizes the media exposure and narrative reframing generated by global broadcasts, which

¹ Refers to a possession that its owner cannot dispose of without extreme difficulty.

amplify the host's international profile. Hosts with weaker initial reputations (such as South Africa's 2010 World Cup or Beijing's 2008 Summer Olympics) often use mega-events explicitly to demonstrate organizational competence and infrastructure modernization.

South Africa, whose football team was suspended by FIFA in 1976 following the violent Soweto uprising, hoped that hosting the FIFA World Cup would act as a uniter for the apartheid-stricken nation. "The journey that we traveled to achieve our freedom and democracy is marked by the struggles and sacrifices of many South Africans," said President Jacob Zuma. "The nation is celebrating like never before. South Africa has come alive, and will never be the same again after this World Cup." South Africa spent directly over four billion dollars to host the World Cup, and billions more in corresponding infrastructure improvements in host cities. In all, the government spent over R13 billion on infrastructure and transportation improvements. This spending directly created 130,000 construction jobs, and the World Cup as a whole indirectly resulted in a gain of 415,000 jobs. Ultimately, the World Cup singlehandedly added 0.5%—or R93 billion—to South Africa's GDP in 2010.

2.2 Capital Misallocation Theory

Mega-event hosting frequently produces large and visible investments that are ultimately misallocated relative to other alternative investments. There are three theoretical explanations for this pattern cluster.

First, the political and economic motives for such a large upfront capital investment systematically lead to over-investment. Hosting mega-events like the FIFA World Cup generates political benefits through international prestige and voter goodwill for incumbents, while the initial expenditures are usually paid for by the nation's taxpayer base. Additionally, due to the significance of these political gains, politicians are incentivized to prioritize grand, visible, and ambitious short-term projects like mega-events over less flashy but more forward-looking public goods such as education or health initiatives.

Second, there is an optimism towards inaccurate project appraisals. Promoters and consultants routinely present overly exaggerated demand forecasts and expected returns in order to obtain approval; every Olympic event since 1960 has run over budget. The host-selection process, essentially functioning as an auction, generally leads to awarding the bidder whose most optimistic (and therefore inaccurate) projections are the highest.

Third, in the context of emerging markets, issues arise concerning procurement, corruption, limited fiscal transparency, and soft budget constraints. While these issues are not limited to only emerging markets, it is these developing nations where poor project selection and chasing international prestige leads to a misallocation of capital towards

specialized assets (such as stadiums) with limited post-event use instead of focusing on productive long-term use infrastructure. The Arena da Amazônia, for example, was a \$300 million stadium built in the Brazilian city of Manaus ahead of the 2014 World Cup. The over 40,000-seater stadium hosted just four World Cup matches, and its remote location in the Amazon rainforest has resulted in limited post-event use.

Research on the 2010 South Africa FIFA World Cup found clear short-run gains in international visibility and a temporary rise in tourist arrivals; 309,554 foreign tourists arrived specifically for the World Cup (although notably lower than the estimate of 450,000–500,000 arrivals). However, its long-run economic legacy is mixed: while certain transport and airport upgrades generated durable capacity increases in host metros, several stadiums have underperformed as revenue-generating assets and pose substantial maintenance liabilities. Major airport upgrades in Durban and Cape Town as well as the introduction of the Gautrain Rapid Rail Link (connecting Johannesburg and Pretoria) certainly represent long-term use; the \$600 million Cape Town Stadium struggled to create profits following the World Cup, losing up to \$8 million annually in the years following 2010. Researchers emphasize modest or negligible sustained FDI gains attributable to the Cup and highlight concerns over the opportunity cost of public funds redirected toward event infrastructure rather than social services.

The existing literature on Brazil's 2014 World Cup emphasizes similar issues with massive public spending and a lack of actual long-term economic impact in favor of localized and corrupt short-term infrastructure construction. Auditors unearthed \$275 million in alleged price-gouging after examining the Brasília stadium project. The need for host cities to declare holidays on game days to ease traffic congestion cost an estimated \$5 billion in lost productivity, and several new stadiums (such as the aforementioned Arena da Amazônia) were constructed in cities without major professional football teams. Many of the planned bus and rail lines were either cancelled or remain unfinished, despite an initial planned investment of \$6.6 billion. Visibly, there were significant concerns over the forced evictions of communities, which sparked mass protests domestically in 2013. Scholars conclude that benefits were uneven and often outweighed, in political terms, by reputational and distributive costs in affected municipalities.

Research on Qatar's 2022 World Cup highlights a very large scale of upfront investment across stadiums, transport, hospitality, and urban redevelopment. Specifically, Qatar planned around its stadiums to ensure a more productive post-event legacy, and many stadiums were modular and were designed to have their upper tiers dismantled to downsize to a more appropriate capacity. Stadium 974, which was built entirely out of shipping containers, was designed to be completely disassembled, with the space later intended

to be used for schools and community facilities. The Doha Metro System, introduced nine years after Qatar won their World Cup bid in 2010, was built primarily to handle mass transit for the World Cup and today remains a core public transport system for Doha. The country's largest airport, Hamad International, also received expansion upgrades to accommodate the influx of tourists and travelers. However, most literature focuses more on the social costs of the tournament; one of the most controversial topics leading up to the event was Qatar's use of the Kafala sponsorship system, which trapped the predominantly South Asian workforce into inappropriate working conditions and wage theft.

2.3 Research Gaps

Most of the existing literature evaluates these mega-events individually and focuses on the hosts in isolation. There has been considerably less attention paid towards the comparative analyses of recent emerging-market hosts that pits the two hypotheses (capital misallocation versus investment signaling) against each other. This paper aims to fill that gap by conducting a longitudinal, cross-country comparison of three emerging World Cup hosts: South Africa (2010), Brazil (2014), and Qatar (2022).

3. Methodology

This paper uses a comparative, longitudinal pre-post analysis to evaluate the macroeconomic impact of hosting the World Cup in three different emerging economies. South Africa (2010), Brazil (2014), and Qatar (2022) were selected as the most recent emerging economies to host the event. Other host countries were not considered—South Korea (2002) was omitted due to their joint-host status with Japan that year, and Russia (2018) was omitted as unusual factors like sanctions, capital flight, and inconsistent data reporting would complicate isolating the effects of hosting the World Cup. Germany (2006), France (1998), United States (1994), and Italy (1990) were not considered due to their already developed status.

Key economic indicators will be considered to evaluate the impact of World Cup hosting; annual GDP growth, FDI as a percentage of GDP, general government debt as a percentage of GDP, and international tourism trends are a non-exhaustive list of metrics that will be considered. A 15-year time period will be implemented to evaluate the before ($T - 7$) and after ($T + 7$) the event.² Host countries will not be compared against non-host countries due to the presence of several uncontrollable variables that would make it difficult to represent the isolated effects of World Cup hosting. The quantitative data for each country's macroeconomic trends will be interpreted using qualitative and case-specific sources derived from prior research.

²Qatar's lack of post-event data is a key limitation of this study.

4. Findings and Analysis

4.1 Case Study 1: South Africa (2010)

South Africa experienced a clear short-run boost in inbound tourism associated with the 2010 World Cup, with the official tourism board reporting a rise in arrivals from roughly 7.0 million in 2009 to 8.1 million in 2010, and a smaller increase to 8.34 million in 2011 (a 3.3% year-over-year increase). These figures reflect the World Cup bump of about 309,000 visitors who came primarily for the event. Their FDI inflows were varied around 2010, and there is no persistent or sustained step-change that can be cleanly attributed to hosting the World Cup.

The qualitative evidence from du Plessis and Maennig (2011) shows that while the 2010 World Cup was a major reputational success, its direct economic impact was far smaller than expected. The tournament certainly enhanced South Africa's global image and generated large volumes of positive international media coverage, but the actual increase in foreign arrivals during the event was much lower than government projections, in part because many regular tourists avoided the country due to anticipated congestion and high prices. As a result, the net tourism gain (which was the key economic channel expected to justify the event) was modest instead of transformative.

Simultaneously, the study highlights structural issues more aligned with the capital misallocation theory. The tournament required substantial public spending on stadiums and infrastructure, but these investments lacked the long-term draw needed to generate sustained economic returns, eventually becoming white elephant investments today. The R30 billion (approximately \$3.6 billion USD today) in direct government spending for the event saw constant overruns, and the 415,000 created jobs was largely temporary and their benefits were not sustained. Overall, it is suggested that the World Cup's legacy leaned more toward a temporary reputational uplift rather than a durable economic stimulus.

While the trends in international visitors to South Africa were generally positive, they have not been strong. Their FDI follows a similar trend. Several stadiums have low utilization and require subsidized upkeep, and some municipalities faced fiscal strains and reallocated public budgets toward maintenance and debt service, all of which are classic symptoms of capital misallocation.

4.2 Case Study 2: Brazil (2014)

Brazil similarly experienced a short-run increase in tourism volumes before and during the 2014 World Cup, but the effect was smaller and more uneven than government projections. The Ministry of Tourism reported a rise to roughly 6.4 million international arrivals in 2014, reflecting an event-year bump but not a significant break in the long-run trend. As Baade and Matheson (2016) note, Brazil's tourism patterns before and after the World Cup show no clear, sustained deviation from their pre-event trajectory, and the anticipated

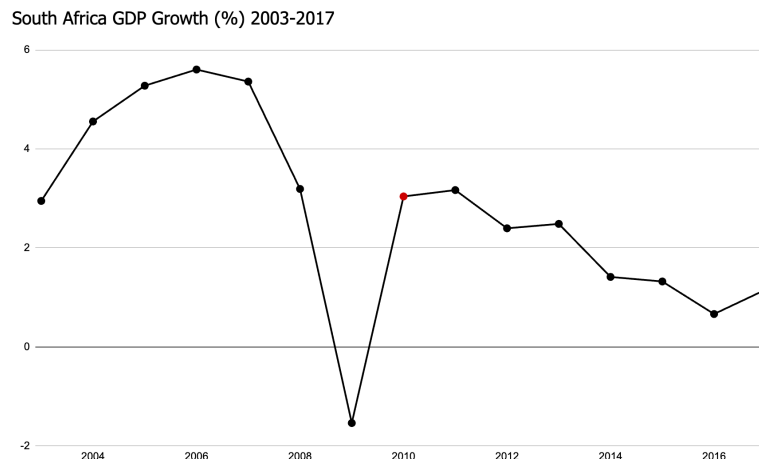


Figure 1. South Africa annual GDP growth (%), 2003–2017.

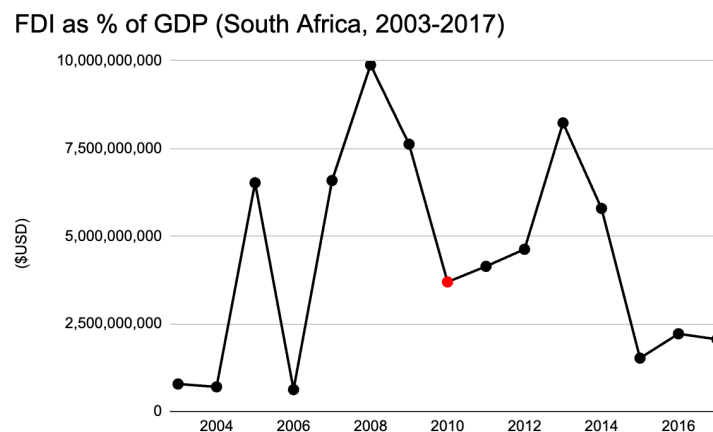


Figure 2. South Africa's FDI, 2003–2017.

surge in repeat visitors did not materialize in the following years. FDI inflows in the early 2010s were also volatile due to macroeconomic and commodity-sector factors, and studies such as Zimbalist (2015) emphasize that no persistent investment uplift can be attributed specifically to World Cup hosting.

Qualitative studies highlight similar patterns of smaller-than-expected economic returns paired with significant political and economic distortions. Zimbalist (2015) again argues that the Brazilian government's pre-event forecasts systematically overstated the long-term effects of the tournament (similar to South Africa), while Gaffney (2014) found that tourist spending and localized service sector gains did not offset the scale of public investment. The event elevated Brazil's global visibility and temporarily improved international perceptions, but these reputation gains were short-lived and rapidly overshadowed by macroeconomic slowdown and political crisis.

At the same time, the Brazil case strongly reflects the capital misallocation mechanism. Research documents severe cost overruns, opaque procurement processes, and politically motivated siting decisions; particularly for stadiums in cities without strong football markets, such as Brasília, Manaus, and Cuiabá. Müller (2015) and Bienen and Gitmez (2016) show that several of these venues quickly became traditional white elephants, operating at low utilization rates and requiring continuous public subsidies. Municipal and state governments faced fiscal strain and rising debt burdens, driven both by stadium maintenance and by underused transport projects initially justified through inflated demand projections. Taken together, the post-event evidence suggests that, while Brazil experienced some short-term signaling effects, the long-run legacy overwhelmingly aligns with the capital misallocation and winner's-curse frameworks, with large, expensive, politically driven projects yielding limited durable economic benefit.

International Tourism, Number of Arrivals, South Africa
2004-2016

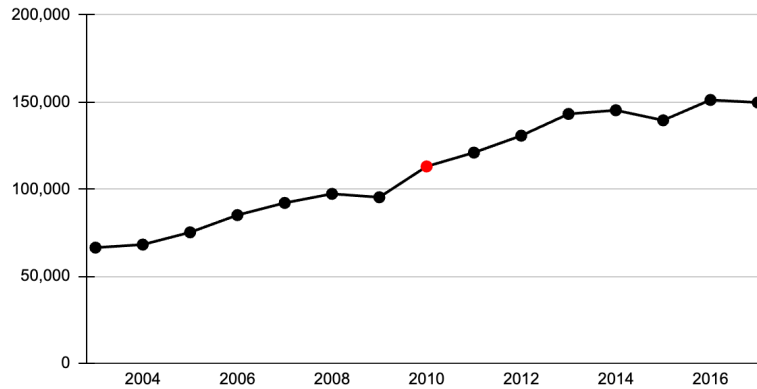


Figure 3. South Africa's international tourism trends.

Brazil GDP Growth (%) 2007-2021

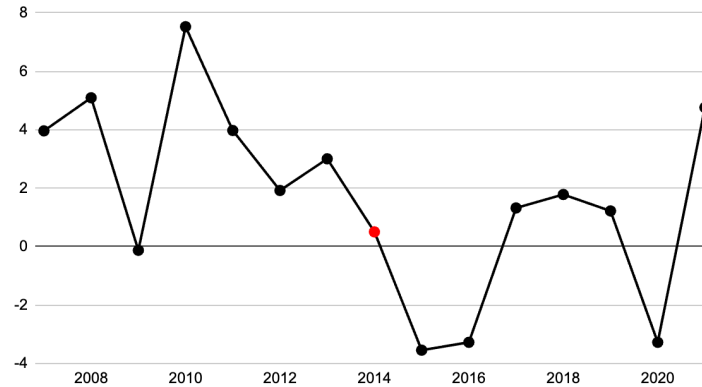


Figure 4. Brazil annual GDP growth (%), 2007–2021.

4.3 Case Study 3: Qatar (2022)

Qatar experienced a significant short-term tourism and visitation spike during the 2022 World Cup, driven by both eventgoers and an unprecedented increase of short-stay travelers. Official figures reported 1.05 million visitors in November–December 2022, reflecting the enormous scale of the tournament relative to Qatar's small population base. However, studies such as Reiche (2023) and Owen (2022) note that this spike was highly concentrated in the event window, with post-tournament visitation returning quickly toward pre-event baselines. The same pattern applies to foreign investment: while Qatar saw large investment inflows into construction, hospitality, and logistical capacity in the lead-up to 2022, these were primarily state-driven capital expenditures rather than market-driven FDI. As Amara (2023) emphasizes, there is no clear evidence of a sustained post-event FDI acceleration, suggesting that the World Cup did not fundamentally reshape Qatar's investment trajectory.

Qualitative research finds that Qatar achieved substantial reputational visibility, aligning with the investment signaling argument. The event projected the state as globally connected, technologically advanced, and capable of delivering a complex mega-event on schedule; a narrative emphasized by Brannagan and Giulianotti (2018) and Dalziel and Samaranch (2022). However, these gains were tempered by the media focus on labor and human rights controversies, which researchers argue damaged the reputational uplift the Qatari government sought.

At the same time, Qatar's case strongly illustrates the core dynamics of the capital misallocation framework. Researchers such as Owen (2022) and Henderson (2023) document that Qatar spent over \$200 billion on infrastructure, much of it unrelated to the stadiums themselves but justified through the event. Many of the new stadiums have limited long-term demand, and although some have been downsized for post-event use, their maintenance and repurposing costs

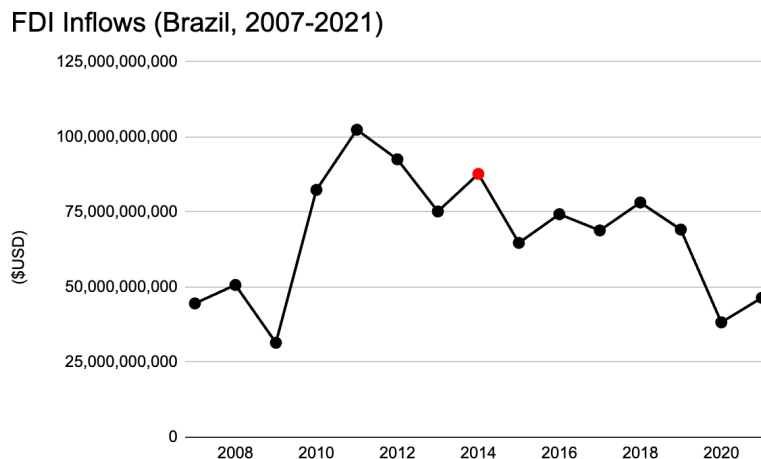


Figure 5. Brazil FDI as % of GDP, 2007–2021.

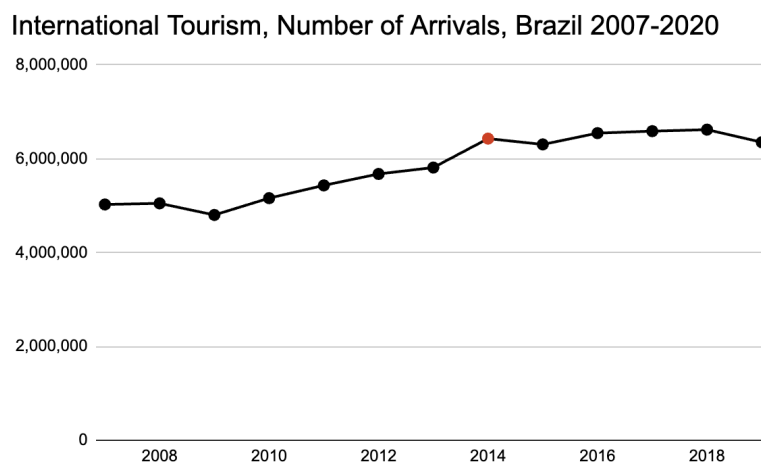


Figure 6. Brazil's international tourism trends, 2007–2018.

remain substantial. Amara (2023) and Reiche (2023) both argue that these projects risk becoming white elephants, due to Qatar's modest domestic sports market and limited year-round events capable of filling large venues. Moreover, the extreme scale of public expenditure (which was financed directly by the state rather than FDI) raises questions about opportunity costs, especially in non-hydrocarbon sectors the government claims to prioritize.

Overall, while Qatar successfully used the World Cup to assert itself on the global stage and demonstrate governance capacity, the economic legacy is far more ambiguous. The evidence suggests a sharp but temporary signaling effect overshadowed by structurally unproductive, state-led capital spending whose long-term returns remain uncertain. The case aligns more closely with capital misallocation than with durable signaling-induced economic gains, but it is too soon to tell.

5. Conclusion

This paper examined whether hosting the FIFA World Cup serves as a credible investment signal for emerging markets or instead reflects the dynamics of capital misallocation and the winner's curse. Across South Africa (2010), Brazil (2014), and Qatar (2022), the evidence points overwhelmingly toward the latter. Each host experienced a short-term boost in tourism and global visibility, but these gains were modest, temporary, and insufficient to generate sustained growth in foreign direct investment or long-run structural benefits. By contrast, all three countries incurred substantial public costs, constructed specialized venues with limited post-event use, and faced long-term maintenance or fiscal burdens; patterns consistent with the systematic overoptimism and politically driven overspending identified in the megaproject literature.

Ultimately, while the World Cup can showcase national

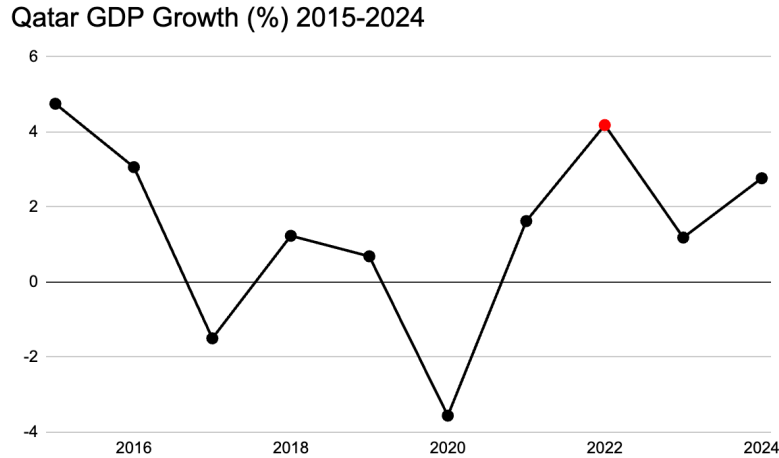


Figure 7. Qatar's GDP growth (%), 2015–2024.

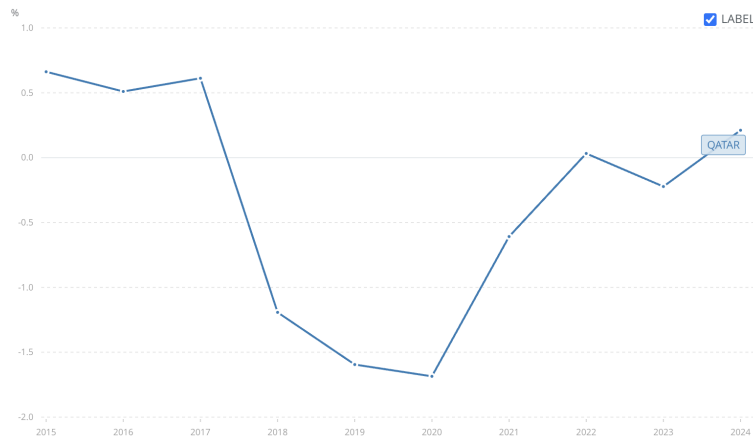


Figure 8. Qatar's FDI as % of GDP, 2015–2024.

capacity and produce a transient reputational uplift, it rarely functions as a durable economic catalyst for emerging markets. Instead, the event's legacy tends to be shaped by large public outlays, underutilized infrastructure, and opportunity costs that outlast the tournament itself. For future hosts, the central lesson is clear: mega-events generate reliable political spectacle but uncertain economic returns. Without stringent cost controls, legacy planning, and alignment with broader development goals, the World Cup risks becoming less a signal of national strength than a showcase of distorted investment priorities.

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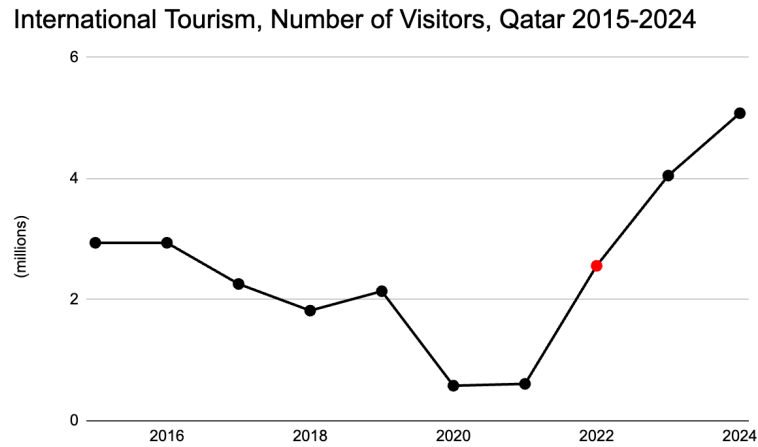


Figure 9. Qatar’s international tourism trends, 2008–2023.

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Singapore Case Study: The Model AI Governance Framework

Amanda Palmar

Abstract. Singapore's Model AI Governance Framework (2020) is examined as an instance of regulation used as an instrument of economic strategy rather than as a constraint on it. The paper traces the framework's principles and operational measures, its implementation through the National AI Strategy and programmes such as 100 Experiments and the AI Apprenticeship Programme, and the associated outcomes in start-up formation, venture capital inflows, and human capital. It argues that Singapore competes on regulatory credibility rather than on scale or cost, and that this credibility has itself become a tradeable economic and diplomatic asset.

1. Country Summary

Singapore has consistently ranked among the world's most advanced digital economies, leveraging its compact geography, strong institutions, and high-income status to lead in artificial intelligence (AI) readiness. As of 2024, its gross domestic product (GDP) stands at US \$547.39 billion (shown in Figure 1) and the digital economy contributes roughly 17% of GDP, reflecting the country's successful transformation into a global hub for technology and services.

Singapore's economic rise has been driven by a deliberate strategy of investing in digital infrastructure and building a skilled workforce. This has produced results: Singapore ranked first in Oxford Insights' Government AI Readiness Index in 2019 and continues to hold a leadership position globally.

Population trends and education have played a key role. Unlike emerging economies, Singapore's development is driven less by rapid population growth and more by strategic investment in human capital, reflected in its Human Capital Index score of 0.88, one of the highest globally. As a developed economy bridging East and West, Singapore exemplifies how advanced nations can leverage AI adoption to enhance productivity through skill transformation.

2. Policy: The Model AI Governance Framework (PDPC 2020)

Singapore issued the Model AI Governance Framework (2020) through the Personal Data Protection Commission (PDPC) to embed trust in AI systems. Its two guiding principles are:

1. *Explainability, transparency, and fairness*: ensuring AI decisions are interpretable and auditable.
2. *Human-centric AI*: keeping humans "in the loop" where AI decisions affect individuals.

These translate into four operational measures:

- (a) internal governance structures (top-level sponsorship and accountability);
- (b) defined human involvement in decision-making;
- (c) robust operational management (data governance, procurement, skills);
- (d) stakeholder communication (transparent public engagement).

The framework is sector-agnostic and aligns with international ethics standards set by the Organisation for Economic Co-operation and Development (OECD), the World Economic Forum (WEF), and the Global Partnership on Artificial Intelligence (GPAI), positioning Singapore's approach as collaborative and exportable.

3. Implementation

The framework underpins Singapore's National AI Strategy (NAIS), which targets five high-value sectors: healthcare, education, smart estates, safety and security, and transport and logistics. To operationalize this, the government launched the Call for Solutions mechanism, sourcing information and communications technology (ICT) and AI innovations through staged pilots that reduce risk and crowd in private suppliers. The framework is operationalized through the following programs:

- (a) *100 Experiments (100E)*: by 2021, generated \$58 million in added value across healthcare, logistics, and finance.
- (b) *AI Apprenticeship Programme (AIAP)*: hands-on training producing Chartered AI Engineers.
- (c) *AI4E, AI4S, and AI4I*: public, student, and industry-targeted literacy programs that together trained more than 70,000 people by 2021, reducing skill bottlenecks.

4. Start-up Support and Development Outcomes

4.1 Whole-Economy Impact

Singapore's National AI Strategy 2.0 (NAIS 2.0) is a policy response to long-term labour and productivity constraints rather than population-driven expansion. The government has invested over S\$500 million in AI research and development through AI Singapore (AISG) under the Research, Innovation and Enterprise (RIE) 2020 and 2025 plans, signaling a structural commitment to embed innovation across the economy.

This ecosystem now includes over 1,100 AI start-ups, 150 AI research and product teams, and 80 research faculty working in Singapore, reflecting a concentrated base of human and technological capital. Complementary initiatives like the Call for Solutions Initiative, AI Trailblazers,

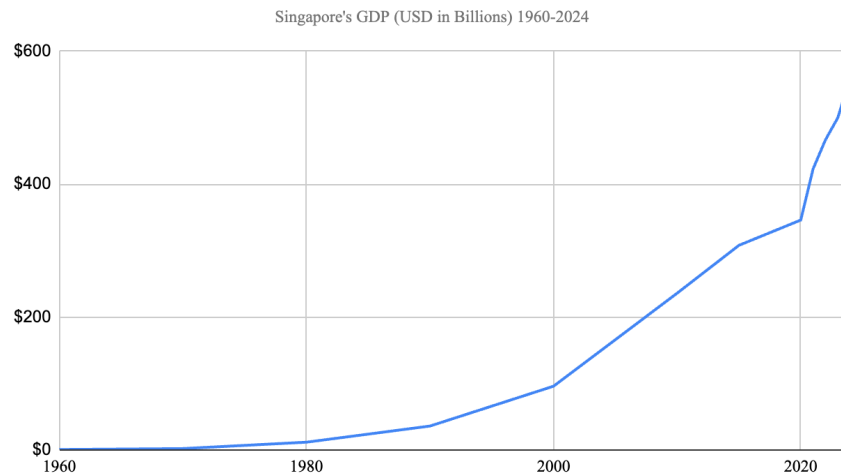


Figure 1. Singapore's gross domestic product (USD in billions), 1960–2024.

Source: EMI Research Team based on World Bank data, <https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=SG>. Accessed October 2025.

and 100E provide financial and infrastructural support for experimentation between research institutions, government agencies, and the private sector.

Singapore's innovation model emphasizes trust and global leadership: it ranks first in the Stanford Global AI Vibrancy Index (per capita), second in the Oxford Insights Government AI Index, and third in the Tortoise Global AI Index. These coordinated efforts have not only strengthened Singapore's innovation capacity but also accelerated measurable economic growth within the AI sector. As shown in Figure 2, the market size of AI in Singapore is projected to expand sharply between 2020 and 2031, reflecting the outcomes of this policy-enabled innovation ecosystem.

4.2 Industry-Level Impact

At the sectoral level, NAIS 2.0 directs AI adoption toward both leading economic sectors (manufacturing, financial services, transport and logistics) and biomedical sciences and Smart Nation priorities such as healthcare and education. These industries anchor Singapore's shift toward high-value, knowledge-intensive growth.

The government supports AI Centres of Excellence (CoEs) and is developing sectoral CoEs with major firms to drive research and adoption within each domain. For example, American Express's Decision Science CoE in Singapore uses AI for credit risk, fraud detection, and customer marketing optimization. Similarly, start-up accelerator programs and the 100E collaboration framework allow private and public organizations to co-develop AI solutions for real-world problems, from predictive maintenance in manufacturing to medical diagnostics.

This structure is reinforced by human capital policy: Singapore aims to train 15,000 AI practitioners through reskilling and professional programs, ensuring that work-

force development keeps pace with industrial transformation. By combining targeted investment, coordinated policy design, and international collaboration, Singapore's innovation ecosystem transforms AI into a cross-sector growth engine that strengthens both its domestic economy and global competitiveness.

In line with this strategic direction, recent data highlight a significant trend in Singapore's AI start-up landscape. According to Statista, the most funded AI start-ups in Singapore include Trax Technology Solutions (retail and computer vision), Carro (automotive and mobility), Advanced Technology Ltd (financial services), Bolttech Digital Brokerage Singapore Pte (insurance), and PropertyGuru Pte (real estate). These start-ups exemplify Singapore's shift toward sector-specific AI applications, demonstrating how targeted policies foster innovation across diverse industries, from mobility and finance to retail and real estate.

4.3 Development Outcomes

The start-up ecosystem now thrives on accountability as a competitive advantage. Singapore's model has shown that regulation does not suppress innovation. On the contrary, it mitigates the risk of experimentation, enabling start-ups to scale responsibly.

5. Economic Impacts

5.1 Human Capital as Economic Capital

Despite rapid innovation and AI deployment, Singapore maintains a very low unemployment rate (approximately 3.2% in 2024) according to World Bank data, suggesting that the policy-enabled AI ecosystem is complementing human labour rather than displacing it. The programs bridge skill gaps, extending long-term productivity. This extensive human capital investment supports Singapore's

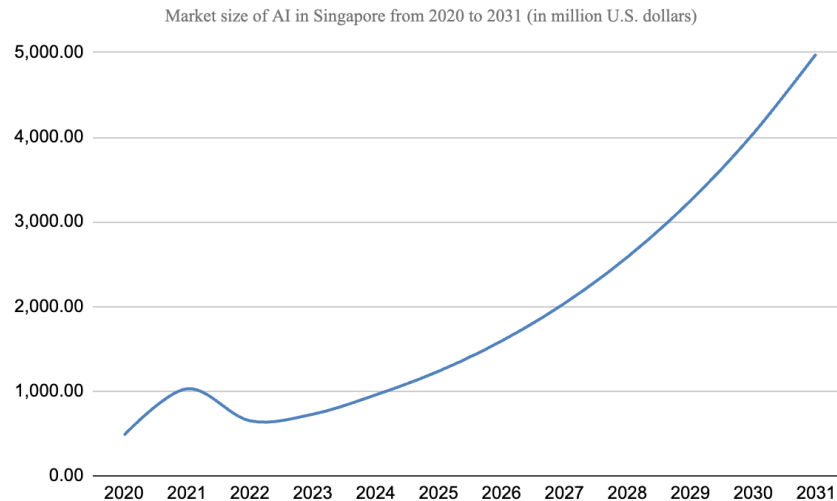


Figure 2. Market size of AI in Singapore, 2020–2031 (millions of U.S. dollars).

Source: EMI Research Team based on Statista, <https://www.statista.com/statistics/1551640/singapore-ai-market-size/>. Accessed October 2025.

Human Capital Index score of 0.88, one of the highest globally, ensuring adaptability in a rapidly evolving digital economy.

5.2 Venture Capital and Private Investment

Transparent data governance and consistent regulatory oversight have turned Singapore into a preferred destination for venture capital funding. As shown in Figure 3, venture capital investment in AI has grown since 2014, including through global downturns, indicating that Singapore's trust-based governance translates into sustained investor confidence even during periods of global market volatility.

This stability reflects how transparent governance and government-led pilot schemes, such as Call for Solutions and AI Singapore's 100E, serve as signal mechanisms for private capital, assuring that projects meet reliability and compliance benchmarks. The presence of strong institutional trust allows start-ups to attract both domestic and international investors seeking long-term stability and policy predictability.

5.3 International Competitiveness

Singapore's leadership in AI readiness and adoption has influenced frameworks elsewhere, including the bilateral arrangement with the European Union on AI safety and cooperation, the OECD Principles (2024), and World Economic Forum initiatives, showcasing governance as a form of soft economic power. In this way Singapore converts policy credibility into international partnerships, reinforcing its position as a trusted hub for cross-border AI collaboration and sustainable innovation.

6. Global Significance

Singapore's approach to artificial intelligence governance has evolved into a form of geoeconomics, visible at the inter-

section between regulatory innovation and diplomacy. Its leadership is most evident in how it uses policy to influence international cooperation and market access. Rather than competing on scale or cost, Singapore competes on credibility. The country offers a trusted environment for venture capital investment, AI testing, and digital trade agreements. This has positioned it as a connector between global technology industries, linking Western regulatory frameworks with emerging Asian markets.

For other nations, Singapore's experience underscores that competitiveness in the AI age will not depend solely on data volume or hardware capacity, but on the policy framework that sustains innovation and human capital investment over time. It offers a template for long-term legitimacy in digital transformation, in which ethics itself becomes a source of economic and diplomatic power.

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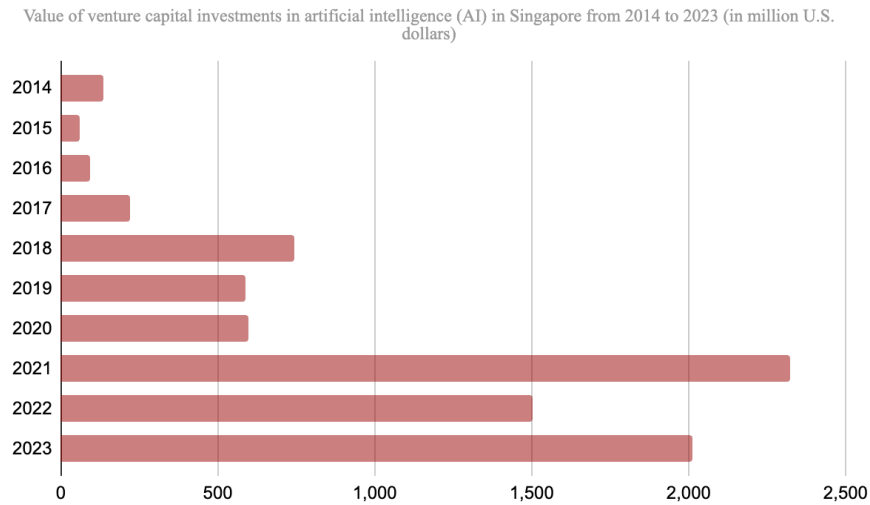


Figure 3. Value of venture capital investments in AI in Singapore, 2014–2023 (millions of U.S. dollars).

Source: EMI Research Team based on Statista, <https://www.statista.com/statistics/1391741/singapore-venture-capital-investment-value-in-ai/>. Accessed October 2025.

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How Mexican Fintech Lenders Are Expanding Access to Capital: A Case Study of Konfío

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Abstract. Mexico hosts Latin America’s second-largest fintech ecosystem, yet roughly half of its adults remain outside the formal banking system and its small and medium-sized enterprises face a credit gap on the order of \$164 billion. This paper examines Konfío, a Mexico City digital lender that underwrites on alternative data—principally electronic tax invoices—rather than collateral or credit history. It finds that Konfío has extended credit at national scale and counter-cyclically, but that high funding costs, repayment volatility, uneven digital access, and limited algorithmic transparency bound what digital lending alone can accomplish.

1. Introduction

Over the past decade, Mexico has become one of Latin America’s most dynamic fintech markets. Hundreds of digital lenders, payment platforms, and financial apps now operate across the country, each promising faster and more accessible financial services. Yet despite this growth, financial exclusion remains widespread. Recent national financial inclusion reports and Global Findex data show that only about 49 percent of adults in Mexico have a formal bank account, which means nearly 66 million people remain unbanked or underbanked (Megginson, 2024). This gap has created a landscape where digital innovation is expanding rapidly but structural barriers continue to limit inclusion.

These challenges are especially visible among small and medium enterprises (SMEs). SMEs represent 99 percent of firms and contribute more than half of the national GDP, yet they face a credit gap of roughly \$164 billion (*The Truth about Mexico’s Financial Exclusion—and How Fintech Is Fixing It*, 2025). Traditional banks often require collateral, long credit histories, and extensive documentation, which many small or informal businesses cannot provide. As a result, a large portion of the SME sector operates without steady access to working capital.

This paper examines how fintech lenders, particularly Konfío, have attempted to expand access to credit for Mexican SMEs, and what limitations or risks remain. It argues that Konfío has opened new pathways to formal financing by using digital onboarding, alternative data, and automated credit scoring to evaluate businesses that banks typically overlook. At the same time, the company operates within a system shaped by informality, unequal digital access, repayment challenges, and concerns about transparency. These factors limit how far fintech alone can close the SME credit gap.

The paper begins by outlining financial exclusion in Mexico and the rapid expansion of its fintech ecosystem. It then presents key findings from the literature on digital financial inclusion and SME finance. The core section provides a qualitative case study of Konfío’s model, scale, and impact. The discussion considers the broader implications of Konfío’s approach and the structural constraints that still shape financial access in Mexico.

2. Financial Exclusion and the Fintech Boom in Mexico

Mexico’s financial system has long been dominated by a small number of large banks that focus on low-risk clients with stable income and strong credit histories. For many small entrepreneurs, the process of applying for a traditional loan is expensive and intimidating. Lack of collateral and formal financial records often leads to automatic rejection, which pushes many SMEs into informal financing networks or forces them to operate without external capital.

At the same time, Mexico has experienced a rapid expansion of its fintech sector. A 2024 ecosystem report finds that Mexico now hosts more than 1,000 fintech providers, including 803 domestic fintech firms (*Mexico’s Fintech Ecosystem Enters Scale-Up Mode*, 2025). This makes Mexico the second largest fintech hub in Latin America. A broader mapping by the Inter-American Development Bank (IDB) identifies more than 3,000 fintechs across Latin America and the Caribbean, with Mexico, Brazil, and Colombia accounting for over half of them (*Fintech in Latin America and the Caribbean*, 2022).

Regulatory reforms have supported this growth. In 2018, Mexico enacted the region’s first comprehensive fintech law, establishing licensing regulations for electronic payment institutions and crowdfunding platforms and creating a regulatory sandbox for experimentation (*Fintech 2025 — Mexico*, 2025). This legal clarity encouraged investment and allowed digital lenders to scale rapidly.

Mexico’s digital infrastructure further strengthens the ecosystem. The mandatory electronic invoicing system operated by the tax authority (Servicio de Administración Tributaria, SAT) produces real-time tax invoices for businesses across the country. These standardized digital records give lenders access to reliable information on sales, cash flow, and financial behavior. This system provides a strong foundation for data-driven lending models like Konfío’s.

3. Digital Financial Inclusion and SME Finance

Research on financial inclusion shows that account ownership alone is not enough to ensure meaningful access. The Global Findex reports that many account holders still rely heavily on cash and conduct most transactions informally. In Mexico, persistent distrust of financial institutions, low financial literacy, and widespread informal employment all contribute to limited use of formal financial services (Megginson, 2024).

Studies by the IDB and the International Monetary Fund highlight the barriers that SMEs face when seeking credit. Many small businesses lack collateral, formal credit histories, or extensive documentation. Because traditional risk models rely heavily on these factors, banks often rule out smaller or informal firms. Digital lenders can address this gap by analyzing alternative data such as SAT invoices, bank flows, and daily sales information to assess creditworthiness (Pinto, 2020).

The literature also notes important risks. Fintech loans often carry higher interest rates because of higher borrower risk and the cost of funding. Concerns about data usage and algorithmic transparency appear frequently. A behavioral study involving Konfío found that about 15 percent of SMEs struggled to make payments on time, although most eventually repaid within about 10 days (*Can Aligning Loan Payments with Income Support Small Businesses in Mexico?*, n.d.). Researchers also emphasize that digital lending tends to reach urban and semiformal businesses rather than rural or fully informal enterprises. These limitations shape the context for evaluating Konfío's impact.

4. Konfío's Model

Konfío was founded in Mexico City in 2013 with the goal of improving access to working capital for SMEs. Instead of relying on physical collateral or lengthy credit histories, Konfío uses alternative data drawn from SAT electronic invoices, bank transaction patterns, sales histories, and credit bureau information (Pinto, 2020). This data-driven model allows the company to evaluate businesses that traditional banks may not be able to assess.

The application process is entirely digital. Business owners connect their financial and tax data through the platform, and Konfío uses automated systems to consolidate information into a credit score. PayPal Mexico, which has partnered with Konfío, reports that this approach allows credit decisions in minutes rather than the weeks or months required for bank evaluations (Bland, 2019). Konfío's early loans were typically around 150,000 pesos and were often used for inventory purchases, payroll, or equipment.

5. Scale and Reach

Konfío has grown into one of Mexico's prominent digital lenders. It has raised about \$700 million in funding and reached a valuation of roughly \$1.3 billion (Mikal Khoso,

2023). More importantly, the company has expanded lending at a national scale. As of late 2024, Konfío had provided more than 26 billion pesos in working capital loans to over 85,000 businesses across the country (*Konfío Secures MX\$7.42 Billion to Boost SME Support in Mexico*, 2024).

During one period, Konfío's SME lending expanded by 68 percent while bank SME lending in Mexico declined by approximately 5 percent (*Konfío Sustainability Report*, 2021). This divergence suggests that Konfío provided credit during a time of retrenchment by traditional lenders and may have helped stabilize business operations during economic downturns.

Qualitative accounts from entrepreneurs highlight the platform's accessibility. Many describe Konfío as the first institution willing to lend to them. They emphasize the speed, clarity, and simplicity of the digital process, contrasting it with the bureaucracy of traditional bank applications.

6. Limitations and Risks

Konfío's approach expands access, but several challenges remain. Repayment difficulties are a key concern. The behavioral study conducted in partnership with Konfío shows that around 15 percent of SMEs struggled to make timely payments, reflecting the volatility of small business cash flows in Mexico (*Can Aligning Loan Payments with Income Support Small Businesses in Mexico?*, n.d.).

Another barrier is cost. Fintech loans in Mexico often carry higher interest rates than bank loans. Media analyses report that these loans can be several times more expensive, which reflects both the risk profile of borrowers and the structure of fintech funding (Suárez, 2024).

Access is also uneven. Konfío's model depends on digital literacy, SAT invoicing, and the ability to connect financial data. Urban and semiformal businesses benefit the most, while rural enterprises or fully cash-based microbusinesses remain difficult to reach. BBVA's SME financial inclusion report highlights persistent digital gaps that limit adoption (Salgado, 2025).

Finally, transparency remains an issue. Many entrepreneurs do not fully understand how algorithms evaluate their data or why credit limits change. These concerns mirror broader global debates about algorithmic risk scoring.

7. Discussion

Konfío's experience illustrates the potential of fintech to widen access to finance for SMEs. The company has delivered more than 26 billion pesos in working capital to over 85,000 businesses and has grown lending during periods when banks reduced their exposure. These achievements support the optimistic view in the literature that digital lenders can fill important gaps in traditional financial systems.

However, the scale of Mexico's SME sector means that fintech adoption still reaches only a fraction of the businesses that need support. More than four million SMEs operate

nationwide, and many remain informal or outside the digital ecosystem. High interest rates, repayment challenges, and uneven access reveal that digital lending operates within structural constraints that technology alone cannot overcome.

These findings suggest that fintech should be seen as one element of a broader strategy for financial inclusion. Digital lenders can reduce key barriers, but long-term progress requires improvements in regulation, financial literacy, digital infrastructure, and support for formalization.

8. Conclusion

Mexico's evolving fintech sector demonstrates both the promise and the limits of digital financial inclusion. Konfío has transformed access to credit for thousands of SMEs by using alternative data, automated scoring, and efficient digital onboarding. In a country where many businesses struggle to qualify for traditional loans, these innovations are significant.

Yet fintech alone cannot close the SME credit gap or resolve the deeper inequalities that shape financial access. High interest rates, repayment difficulties, concerns about transparency, and unequal digital adoption all indicate that broader structural reforms are needed. Policymakers can support progress by strengthening consumer protections and investing in digital and financial infrastructure. Fintech firms, in turn, can focus on tools that promote financial planning and long-term resilience.

Overall, Konfío's case suggests that digital lending can play a meaningful role in expanding access to capital, but sustainable inclusion will require coordinated efforts across regulators, financial institutions, and the SME sector.

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